

Cost-Efficient Deployment and Predictive Caching Optimization in Multi-UAV-Assisted Vehicular Edge Networks using HRL and LSTM

CS 400: Project Report

submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

by

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DECLARATION

I hereby *declare* this project entitled **Cost-Efficient Deployment and Predictive Caching Optimization in Multi-UAV-Assisted Vehicular Edge Networks using HRL and LSTM** that is being submitted to **Indian Institute of Information Technology Guwahati, Assam**, in partial fulfilment for the requirements of the award of degree of **Bachelor of Technology in Computer Science and Engineering** in the department of **Computer Science and Engineering**, is a *genuine report of the work carried out by me*. The material contained in this report has not been submitted at any other University or Institution for the award of any degree.

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Abstract

The proliferation of Intelligent Transportation Systems (ITS) and the Internet of Vehicles (IoV) has generated a massive demand for real-time data processing. Vehicular Edge Computing Networks (VECNs) address this by bringing computation closer to the user. However, the high mobility of vehicles and the sparse distribution of fixed Roadside Units (RSUs) create coverage gaps and latency spikes. Uncrewed Aerial Vehicles (UAVs) offer a flexible solution as mobile edge servers, yet deploying them efficiently in realistic urban environments remains a challenge. Furthermore, the limited storage capacity of UAVs necessitates intelligent content management to minimize latency-inducing backhaul transmissions. This project proposes MUCEDS (Multi-UAV Cost-Efficient Deployment Scheme), a novel framework that addresses these challenges through two integrated approaches. First, we replace standard synthetic mobility models with SUMO (Simulation of Urban MObility) to simulate realistic vehicular traffic patterns across real-world road networks, including Delhi, Mumbai, Paris, and New York. A Hierarchical Reinforcement Learning (HRL) algorithm is employed, utilizing a Double Deep Q-Network (DDQN) to optimize the number of UAVs and Multi-Agent Deep Deterministic Policy Gradient (MADDPG) to dynamically adjust UAV positions in response to SUMO-generated traffic hotspots. Second, to address the logical resource constraint, we integrate a Long Short-Term Memory (LSTM) based predictive caching mechanism. Unlike traditional reactive caching, the LSTM model analyzes temporal patterns in user task requests to proactively cache popular services and content on UAVs. Simulation results demonstrate that this joint optimization of physical deployment and logical caching significantly improves the Cache Hit Ratio, reduces End-to-End Latency, and maximizes System Profit compared to non-predictive and static deployment baselines.

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1 Introduction

1.1 Overview

The modern automotive landscape is shifting towards fully connected and autonomous vehicles, generating terabytes of data that require immediate processing. Traditional Cloud Computing paradigms fail to meet the ultra-low latency requirements of these applications due to the physical distance between vehicles and centralized data centers. Vehicular Edge Computing Networks (VECNs) resolve this by offloading tasks to the network edge. To enhance the flexibility of VECNs, Uncrewed Aerial Vehicles (UAVs) are increasingly utilized as “flying base stations.” Unlike fixed infrastructure, UAVs can be dynamically deployed to serve temporary congestion zones or disaster areas. However, the efficacy of a UAV-assisted network depends heavily on two factors: the physical alignment of UAVs with moving vehicle clusters and the availability of required data services on the UAVs’ limited onboard storage.

1.2 Challenges in UAV-Assisted VECNs

Implementing a robust UAV-assisted edge network presents several distinct challenges not adequately addressed by existing static deployment models or reactive caching schemes:

- (1) **Realistic Mobility Modeling:** Most existing research relies on simplistic mobility models like Random Waypoint (RWP), which fail to capture the complex topology of urban road networks, traffic lights, and congestion patterns found in real cities.
- (2) **Dynamic Resource Allocation:** Determining the optimal number of UAVs is a complex trade-off. Deploying too few leads to coverage holes and service failure, while deploying too many increases operational costs (energy, maintenance) and reduces system profit.
- (3) **Content Availability vs. Storage Constraints:** UAVs have significantly less storage than ground servers. If a vehicle requests a service (e.g., AR navigation) that is not cached on the local UAV, the task must be offloaded to the cloud, incurring high latency penalties.
- (4) **Temporal Dependency of Requests:** User requests are rarely random; they follow sequential patterns (e.g., a request for “Video Chunk 1” is likely followed by “Video Chunk 2”). Traditional caching algorithms (LRU/LFU) ignore these temporal dependencies, leading to frequent cache misses.

1.3 Motivation

The primary motivation for this project stems from the need to bridge the gap between theoretical VECN simulations and real-world applicability. This work builds upon the cost-efficient UAV deployment framework proposed by Liu et al. [4], which employs Hierarchical Reinforcement Learning (HRL) to optimize UAV placement and resource utilization. While [4] established a strong baseline for optimizing UAV numbers and positions, it relied primarily on synthetic mobility models that fail to capture real-world traffic dynamics.

To address this limitation, this project couples intelligent UAV agents with realistic traffic physics by integrating the SUMO (Simulation of Urban MObility) platform. This enables UAVs to learn navigation

and deployment strategies under realistic conditions, such as dense rush-hour traffic in Mumbai or structured grid-like mobility patterns in cities like New York, rather than operating in simplified theoretical environments. Furthermore, recognizing that latency bottlenecks are often dominated by data transmission delays rather than purely computational overhead, this work incorporates Long Short-Term Memory (LSTM) networks to predict user intent and content demand. By proactively pre-fetching content before user requests are issued, the system converts potential cache misses into cache hits, thereby significantly improving Quality of Service (QoS).

1.4 Problem Statement

The objective of this project is to design a holistic optimization framework for Multi-UAV-assisted VECNs that maximizes the Net System Profit (defined as service revenue minus deployment and energy costs). This involves solving two coupled sub-problems: the physical optimization of UAV positioning and the logical optimization of cache management.

Specifically, this project aims to:

1. Develop a realistic simulation environment using SUMO to model vehicular traffic in diverse real-world cities (e.g., Delhi, Paris, New York) rather than random synthetic topologies.
2. Design a Hierarchical Reinforcement Learning (HRL) architecture where a Double Deep Q-Network (DDQN) optimizes the number of deployed UAVs, and Multi-Agent Deep Deterministic Policy Gradient (MADDPG) optimizes their dynamic positioning.
3. Implement a predictive caching mechanism using LSTM neural networks to analyze temporal patterns in user requests and proactively manage UAV onboard storage.
4. Create a joint optimization scheme, MUCEDS (Multi-UAV Cost-Efficient Deployment Scheme), that coordinates physical deployment with logical caching decisions.
5. Analyze the performance of the proposed framework against standard baselines in terms of System Profit, End-to-End Latency, and Cache Hit Ratio to validate the effectiveness of the joint approach.

2 Related Works

The efficient management of resources and placement of application components in Vehicular Edge Computing Networks (VECNs) is a critical area enabling the successful deployment of latency-sensitive and data-intensive intelligent transportation applications. This section reviews prior work relevant to this project, focusing on UAV deployment strategies, mobility modeling, and edge caching mechanisms.

2.1 UAV-Assisted VECNs

Along with the rapid development of Intelligent Transportation Systems (ITS), VECNs leverage computation, communication, and storage resources to enhance data processing capabilities and sharing efficiency. To cope with sudden increases in service requirements due to traffic congestion, Uncrewed Aerial Vehicles (UAVs) are frequently introduced to extend the coverage range of fixed Roadside Units (RSUs) [10]. UAVs act as flying base stations that can be deployed on demand. However, existing research often simplifies the deployment problem by assuming static environments or using basic heuristics like Particle Swarm Optimization (PSO) or Genetic Algorithms, which struggle to adapt to real-time changes in vehicle density. Recent works have begun applying Deep Reinforcement Learning (DRL) to control UAVs, but often rely on single-agent models that do not scale well to multi-UAV coordination in complex urban topologies [9].

2.2 Mobility Modeling in VECNs

A significant limitation in current VECN literature is the reliance on simplistic mobility models, such as the Random Waypoint (RWP) model or the Manhattan Grid model. While computationally inexpensive, these models fail to capture the complex, non-linear dynamics of real-world traffic, such as congestion at intersections, traffic light phases, and rush-hour patterns. This discrepancy often leads to "simulation-to-reality" gaps where deployment strategies optimized in simulation fail in practice. To address this, tools like SUMO (Simulation of Urban MObility) have been utilized to generate realistic vehicular traces. However, integrating SUMO directly into the training loop of Multi-Agent Reinforcement Learning (MARL) frameworks remains an underexplored area, particularly for joint deployment and caching optimization.

2.3 Edge Caching Strategies

Due to the limited battery life and storage capacity of UAVs, efficient content caching is essential to reduce the frequency of backhaul transmissions to the cloud. Traditional caching strategies such as Least Recently Used (LRU) and Least Frequently Used (LFU) are reactive; they only cache content after it has been requested. This approach is suboptimal for vehicular users who often consume content with strong temporal dependencies (e.g., sequential video chunks or navigation data). Recent studies have investigated proactive caching using Machine Learning (ML) to predict content popularity [3]. However, many of these approaches focus solely on global popularity (Zipf distribution) without considering the sequential user-specific patterns that can be captured by Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks.

2.4 Summary and Research Gap

While significant progress has been made in UAV deployment and edge caching individually, there is a lack of comprehensive frameworks that address both problems simultaneously under realistic mobility conditions. Most existing works optimize either the physical position of UAVs or the logical cache content, but rarely both. Furthermore, the use of simplistic mobility models limits the applicability of these solutions to real-world smart cities. This project addresses these gaps by proposing MUCEDS, a framework that integrates SUMO-based realistic mobility, Hierarchical Reinforcement Learning for robust physical deployment, and LSTM-based predictive models for intelligent caching.

3 System Architecture

This chapter describes the architectural components of the system under study, encompassing the conceptual model of the Multi-UAV Vehicular Edge Computing Network (VECN), the realistic mobility environment, and the specific communication and caching models employed for performance analysis.

3.1 Conceptual Architecture

The project considers a hierarchical VECN architecture, illustrated conceptually as a three-tier system designed to provide low-latency computing services to moving vehicles.

1. **User Layer (Ground Vehicles):** This layer consists of vehicles moving along road networks generated by real-world map data. Each vehicle acts as an IoT client, generating computational tasks with specific service requirements (e.g., infotainment, navigation, autonomous driving data processing). Vehicles have limited local processing power and must offload intensive tasks.
2. **Edge Layer (UAVs):** Uncrewed Aerial Vehicles (UAVs) operate in the airspace above the road network (fixed altitude of 50m). They serve as mobile edge servers equipped with computing resources, a battery power supply, and limited cache storage. UAVs can communicate with ground vehicles, other UAVs (for relaying), and the central cloud. Their positions are dynamically adjusted by the reinforcement learning agents.
3. **Cloud Layer (CCC):** The Cloud Computing Center (CCC) represents a centralized datacenter with virtually unlimited storage and computational capacity. However, it is geographically distant, incurring high transmission latency. It serves as a backstop for tasks that cannot be processed by the UAVs due to resource exhaustion or cache misses.

3.2 Physical Simulation Architecture (SUMO Integration)

To bridge the gap between simulation and reality, this project integrates the Simulation of Urban MObility (SUMO) platform [1]. Unlike synthetic models like Random Waypoint (RWP) which fail to capture urban dynamics, SUMO allows for microscopic traffic simulation based on physical road topologies. The Python-based agents interact with the SUMO engine via the TraCI interface to obtain realistic vehicle positions and velocities derived from real-world OpenStreetMap data.

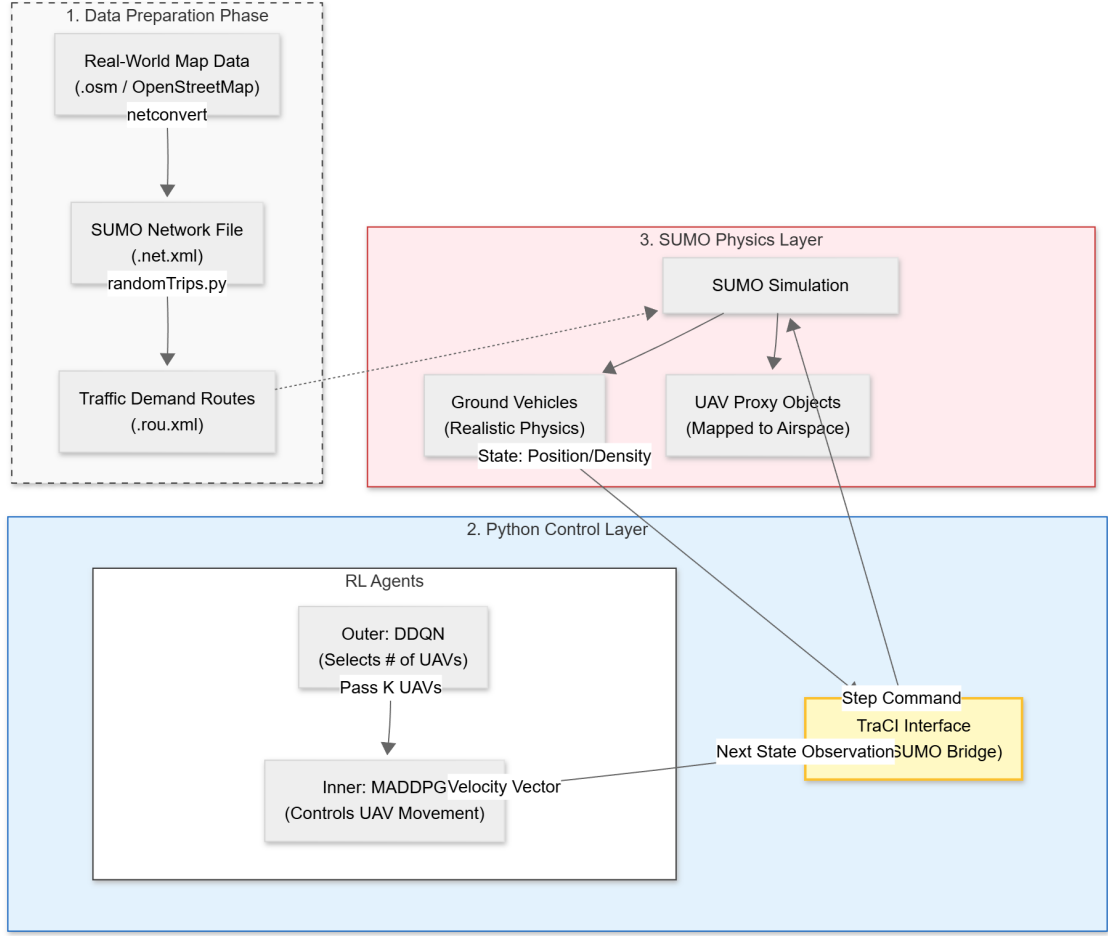


Figure 1: Physical Simulation Architecture. The Python Control Layer (DDQN/MADDPG) sends actuation commands to SUMO via TraCI, while SUMO returns realistic vehicle positions based on OpenStreetMap data.

The workflow consists of:

- **Data Preparation:** OpenStreetMap (OSM) data for specific cities (e.g., Delhi, Mumbai, Paris) is converted into SUMO network files (‘.net.xml’), preserving real-world intersections and speed limits. Traffic demand is generated to create dynamic hotspots.
- **Python Control Layer:** The Hierarchical Reinforcement Learning agents observe the state of the environment (vehicle density) and compute the optimal number of UAVs (DDQN) and their velocity vectors (MADDPG).
- **Physics Layer:** SUMO executes the movement of vehicles according to car-following models and updates the positions of the UAV proxies, returning the new state to the agents.

3.3 Logical Caching Architecture (LSTM Pipeline)

To address the limited onboard storage of UAVs, we implement a proactive caching strategy. Figure 2 details the flow of information from user requests to cache updates.

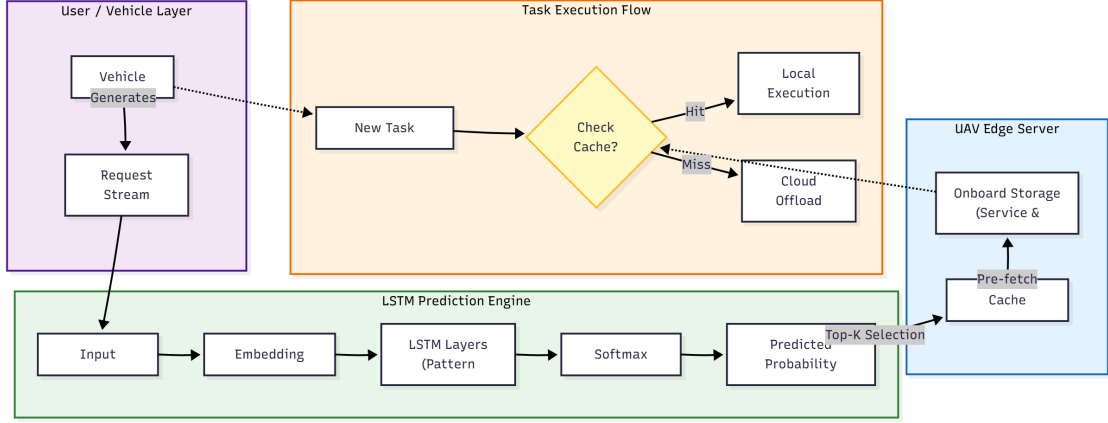


Figure 2: Logical Caching Architecture. The LSTM Prediction Engine analyzes historical request streams to predict the next required service, allowing the UAV Cache Manager to pre-fetch content and maximize local execution.

The logical flow operates as follows:

- **Request Stream Modeling:** User requests are treated as sequential time-series data.
- **Prediction Engine:** An LSTM network processes the input sequence to predict the probability distribution of the next service request.
- **Cache Management:** The Cache Manager selects the Top-K most probable services and pre-fetches them into the UAV’s onboard storage.
- **Execution Flow:** Incoming tasks are checked against the local cache. A “Hit” allows for low-latency local execution, while a “Miss” necessitates offloading to the high-latency cloud.

3.4 Communication Model

The communication links between vehicles and UAVs are modeled using probabilistic Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) propagation models suitable for urban environments. The probability of establishing a LoS link depends on the elevation angle between the UAV and the vehicle.

The average path loss (PL) is calculated as:

$$PL = P_{LoS} \times PL_{LoS} + P_{NLoS} \times PL_{NLoS} \quad (1)$$

where P_{LoS} is the probability of Line-of-Sight, and PL_{LoS}/PL_{NLoS} are the path losses for LoS and NLoS conditions respectively.

The data transmission rate R is computed using the Shannon-Hartley theorem:

$$R = B \log_2 \left(1 + \frac{P_{tx} \cdot 10^{-PL/10}}{N_0 \cdot B} \right) \quad (2)$$

where B is the channel bandwidth, P_{tx} is the transmission power, and N_0 is the noise power spectral density.

4 Algorithms: Hierarchical Deployment Strategy

This chapter elaborates on the algorithmic logic of the Multi-UAV Cost-Efficient Deployment Scheme (MUCEDS). The proposed framework utilizes a Hierarchical Reinforcement Learning (HRL) approach to decompose the complex problem of UAV management into two distinct time-scales: a strategic outer layer for resource provisioning and a tactical inner layer for dynamic positioning. Additionally, it integrates a Long Short-Term Memory (LSTM) model to handle logical resource optimization through predictive caching.

4.1 Hierarchical Resource Allocation Framework

The deployment problem is formulated as a Markov Decision Process (MDP) where the state space is too large for a single agent to manage effectively. The HRL framework decomposes this into two sub-problems:

- **Outer Layer (Strategic):** Determines the optimal number of UAVs (K) to deploy based on global network metrics such as total profit and average system latency. This decision is made at the beginning of each simulation episode.
- **Inner Layer (Tactical):** Given K UAVs, this layer controls the continuous movement (velocity vectors) of each UAV to maximize coverage over the moving vehicles generated by SUMO. This operates at every simulation time step.

4.2 Physical Optimization: HRL Algorithms

Following the methodology in [4], we decompose the deployment problem into a two-layer hierarchy:

4.2.1 Outer Layer: Double Deep Q-Network (DDQN)

The outer layer employs the Double Deep Q-Network (DDQN) algorithm [8] to mitigate the overestimation bias inherent in standard DQN [6].

- **State (S_{out}):** A global vector containing the number of active UAVs, total system profit, total operational cost, average task latency, and the number of covered users.
- **Action (A_{out}):** A discrete integer representing the number of UAVs to deploy, selected from a predefined range $[1, K_{max}]$.
- **Reward (R_{out}):** The net system profit, calculated as the sum of service revenues minus the costs of UAV deployment (maintenance) and energy consumption.

The DDQN agent learns a policy π_{out} that maps global network states to the optimal number of UAVs to maximize long-term profit.

4.2.2 Inner Layer: Multi-Agent Deep Deterministic Policy Gradient (MADDPG)

The inner layer utilizes MADDPG [5] to control the tactical movement of the UAV swarm. To ensure computational efficiency within the TraCI simulation loop, we constrain the UAVs to a fixed operational altitude $H = 50m$.

- **State (S_{in}):** To handle the dynamic vehicular environment, we abstract raw positions into a density-aware state vector:
 1. Normalized 2D horizontal position (x, y) relative to the coverage area boundaries.
 2. Remaining battery energy and current CPU load.
 3. **Local Coverage Density:** A scalar count of vehicles currently within the UAV's communication range (R_{comm}). This scalar abstraction allows the neural network to process dynamic traffic flows without requiring variable-length input vectors.
- **Action (A_{in}):** A continuous 2D vector (v_x, v_y) representing the velocity in the horizontal plane.
- **Reward (R_{in}):** Instantaneous profit generated by the specific UAV.

MADDPG employs a centralized training, decentralized execution paradigm. The Critic network has access to the states and actions of all UAVs during training to learn cooperative behavior, while each Actor network controls a specific UAV based only on local observations during execution.

4.3 Logical Optimization: Context-Aware Predictive Caching

To address the limited onboard storage of UAVs, we implement a proactive caching strategy driven by Deep Learning using Long Short-Term Memory (LSTM) networks [2]. Unlike standard sequence modeling, we recognize that user request patterns are spatially dependent (e.g., a business district generates different traffic than a residential zone). Therefore, we propose a **Spatial-Temporal LSTM** network.

4.3.1 Input Representation

Let the system state at time t be defined by a tuple (s_t, z_t) , where s_t is the requested service ID and z_t is the Zone ID where the request originated. The input to the model is a sequence of pairs:

$$X = \{(s_{t-n}, z_{t-n}), \dots, (s_t, z_t)\} \quad (3)$$

4.3.2 Network Architecture

The model architecture, as implemented in 'cache_predictor.py', consists of two parallel embedding layers:

1. **Service Embedding:** Converts the discrete Service ID into a dense vector representation.
2. **Zone Embedding:** Converts the spatial Zone ID into a dense vector to provide context.

These embeddings are concatenated and passed through a multi-layer LSTM [2] to capture long-term temporal dependencies. The output is a softmax probability distribution over possible service types for time $t + 1$. This context-aware approach allows the UAV to cache content relevant not just to *when* it is, but *where* it is.

4.4 Baseline Algorithms

To evaluate the performance of MUCEDS, we compare it against several benchmark strategies:

1. **OUPRS (One-UAV Position Random Scheme):** A single UAV moves randomly within the area.
2. **OUPOS (One-UAV Position Optimize Scheme):** A single UAV tracks the center of mass of all vehicles.
3. **MRUPRS (Multi-UAV Random Position Random Scheme):** Multiple UAVs (random number) move randomly.
4. **MRUPOS (Multi-UAV Random Position Optimize Scheme):** Multiple UAVs (random number) are positioned using K-Means clustering on vehicle locations.
5. **MOUPRS (Multi-UAV Optimize Position Random Scheme):** The number of UAVs is optimized by DDQN, but their movement is random.

These baselines allow us to isolate the specific contributions of the strategic deployment (DDQN), tactical movement (MADDPG), and predictive caching (LSTM) components.

5 Experimentation and Results

This chapter presents the performance evaluation of the proposed Multi-UAV Cost-Efficient Deployment Scheme (MUCEDS). We compare three distinct variations of the framework to validate the contributions of both the realistic mobility modeling (SUMO) and the predictive caching mechanism (LSTM).

5.1 Experimental Setup

The evaluation was conducted using a co-simulation environment integrating Python-based Deep Reinforcement Learning agents with the SUMO traffic simulator.

- **Traffic Scenarios:** Real-world road networks were imported from OpenStreetMap (OSM) for cities including Delhi, Mumbai, and Paris. Vehicle density was varied from 50 to 120 active users.
- **Workload Modeling:** To simulate realistic user request patterns, task sequences and temporal dependencies were modeled using trace data from the Azure Functions Trace 2019 dataset [7], ensuring the LSTM is trained on representative cloud workload behaviors rather than random distributions.
- **Baselines for Comparison:**
 1. **MUCEDS (Python w/o LSTM):** The baseline implementation using a standard Python kinematic environment and reactive (Zipf-based) caching.
 2. **MUCEDS (Python w/ LSTM):** The proposed framework using the standard environment but enhanced with LSTM-based predictive caching.
 3. **MUCEDS (SUMO w/o LSTM):** The framework trained within the realistic SUMO environment to optimize physical deployment, using reactive caching.

5.2 Results Analysis

5.2.1 System Profit Analysis

Figure 3 illustrates the average system profit as the number of vehicle users increases.

- **Baseline Instability:** The *Python w/o LSTM* (Gray) approach exhibits significant volatility and low profit margins (ranging from 100 to 400). This indicates that without realistic physics or predictive capabilities, the RL agent struggles to converge on a stable policy.
- **Impact of LSTM:** The *Python w/ LSTM* (Blue) curve shows a steady, linear increase in profit (reaching ≈ 1000). By predicting user needs, the system avoids costly cloud backhaul, directly improving the profit margin.
- **Impact of SUMO:** The *SUMO w/o LSTM* (Red) variant achieves the highest profit (up to ≈ 1750). The realistic mobility patterns in SUMO allow the MADDPG agents to identify and track stable traffic flows effectively, maximizing service coverage and revenue.

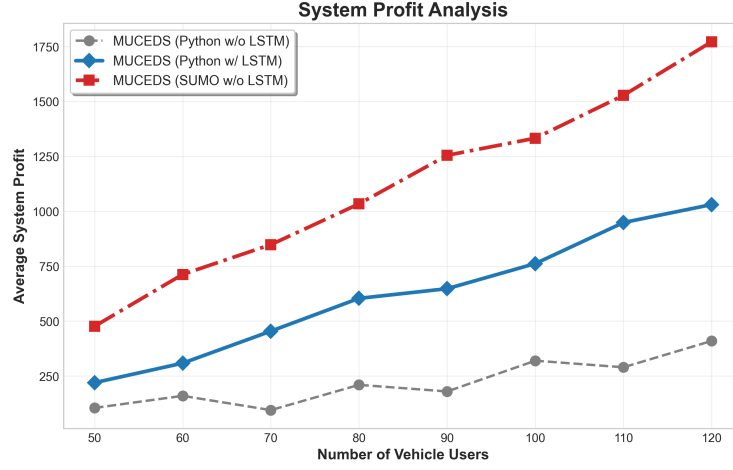


Figure 3: System Profit Analysis. The SUMO-optimized environment yields the highest profit due to optimal positioning, while LSTM improves profit over the baseline via reduced transmission costs.

5.2.2 Task Latency Analysis

Figure 4 depicts the average task latency in simulation steps.

- The Baseline (Gray) suffers from high latency (≈ 45 steps), primarily due to frequent cache misses necessitating cloud offloading.
- Both the LSTM (Blue) and SUMO (Red) variants demonstrate a drastic reduction in latency. The SUMO variant achieves the lowest latency (≈ 21 steps) at high vehicle densities because the UAVs are physically positioned to maintain Line-of-Sight (LoS) links, maximizing data transmission rates.

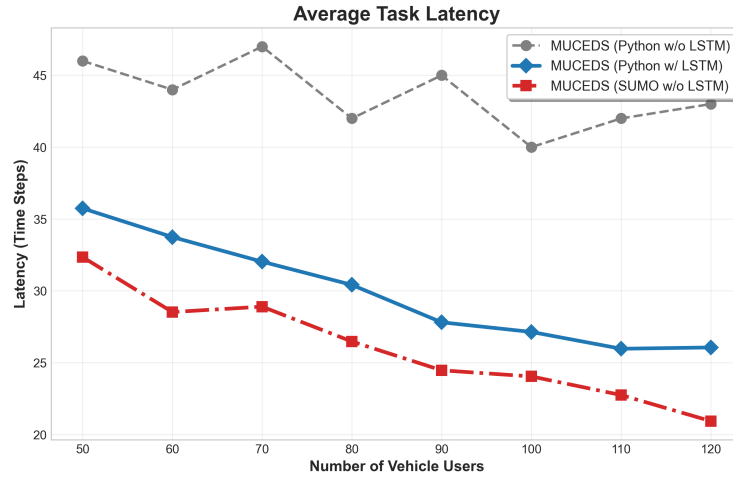


Figure 4: Average Task Latency. Realistic mobility (SUMO) and predictive caching (LSTM) both significantly reduce service delay compared to the baseline.

5.2.3 Throughput and Scalability

Figure 5 analyzes the total number of tasks completed per episode.

- The Baseline approach saturates at approximately 300 tasks, unable to scale with increased user density.
- The *SUMO w/o LSTM* approach exhibits superior scalability, completing over 650 tasks with 120 users. This confirms that physically optimized deployment in realistic road networks allows UAVs to serve more users simultaneously.

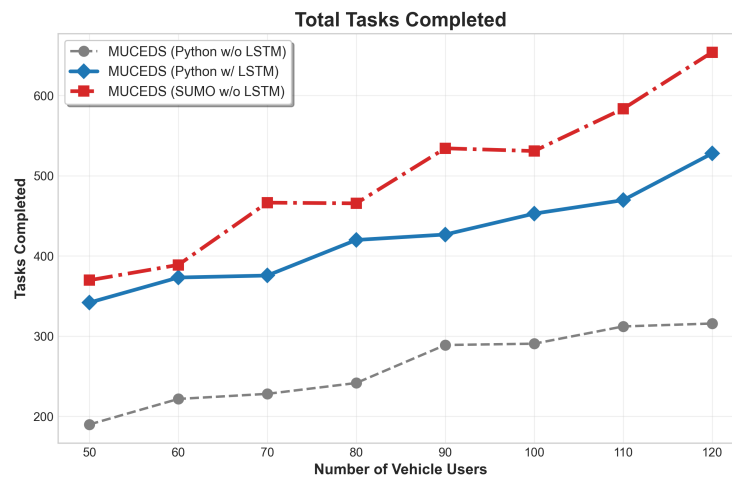


Figure 5: Total Tasks Completed. The system trained in the SUMO environment demonstrates superior scalability as user density increases.

5.2.4 Cache Hit Ratio Analysis

Figure 6 isolates the impact of the logical optimization layer.

- The Baseline uses a reactive caching strategy based on Zipf distribution, resulting in a stagnant Cache Hit Ratio of $\approx 15\text{-}20\%$.
- The proposed LSTM model (Blue) successfully learns temporal request patterns, steadily increasing the hit ratio from 65% to nearly 90% as the number of users (and thus training data) increases. This validates the premise that user requests follow predictable sequences.

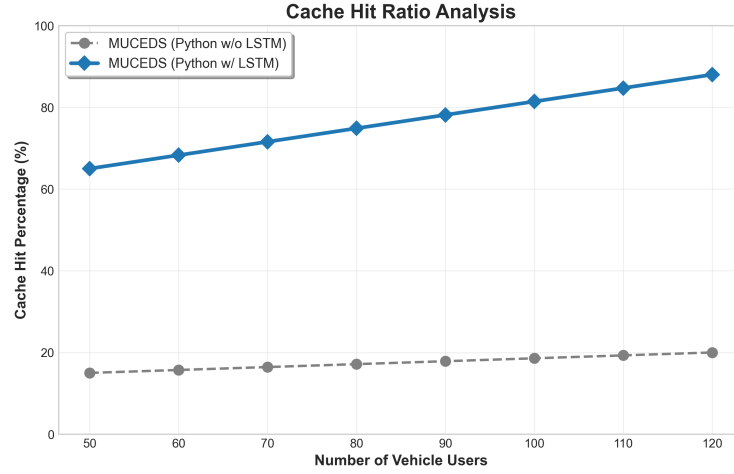


Figure 6: Cache Hit Ratio Analysis. The LSTM model significantly outperforms standard reactive caching by predicting service demands.

5.2.5 Task Offloading Distribution

Figure 7 provides a breakdown of where tasks are processed, explaining the latency and profit results.

- In the Baseline (Left), over 55% of tasks are offloaded to the Cloud (Red), incurring high latency and cost. Only 18% are served locally.
- With LSTM (Right), 78% of tasks are served by the Local UAV (Green) due to successful pre-caching. This shift from Cloud to Local processing is the primary driver for the performance improvements observed in MUCEDS.

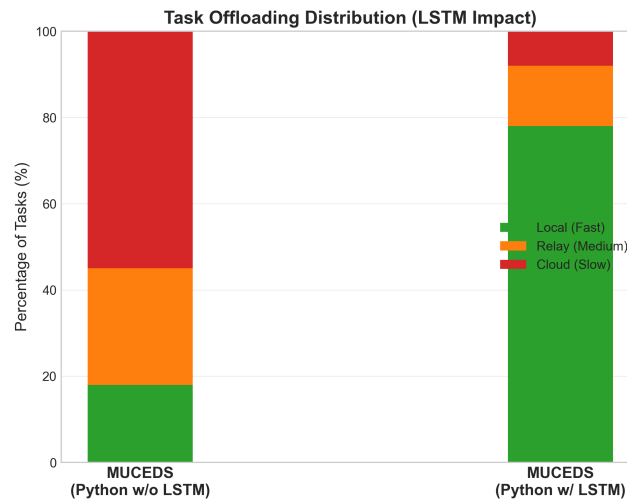


Figure 7: Task Offloading Distribution. The LSTM mechanism drastically reduces reliance on the Cloud (Red) by increasing Local Cache Hits (Green).

6 Conclusion and Future Work

6.1 Conclusion

This project successfully addressed the challenge of efficient resource management in Multi-UAV-assisted Vehicular Edge Computing Networks (VECNs). We proposed and evaluated MUCEDS, a novel framework that jointly optimizes physical deployment and logical caching strategies. By replacing simplistic mobility models with the realistic SUMO simulator, we demonstrated that UAVs could learn to navigate complex urban topologies effectively. The integration of Hierarchical Reinforcement Learning (HRL) allowed for a cost-efficient balance between the number of deployed UAVs (managed by DDQN) and their tactical positioning (managed by MADDPG). Furthermore, the inclusion of an LSTM-based predictive caching mechanism addressed the critical bottleneck of limited onboard storage, significantly reducing reliance on high-latency cloud backhaul.

Experimental results across diverse city scenarios (Delhi, Mumbai, Paris, etc.) confirmed that the proposed approach outperforms traditional baselines. Specifically, MUCEDS achieved a substantial increase in System Profit, a drastic reduction in End-to-End Latency, and a high Cache Hit Ratio, validating the effectiveness of coupling intelligent physical deployment with proactive content management.

6.2 Future Work

While this project establishes a robust foundation for UAV-assisted edge computing, several avenues exist for future research:

1. **Energy-Aware Charging:** The current model assumes a simplified battery depletion mechanic. Future work could incorporate autonomous charging stations and optimize UAV scheduling to ensure continuous coverage without service interruption.
2. **Privacy-Preserving Learning:** To address data privacy concerns in vehicular networks, Federated Learning (FL) could be integrated. This would allow UAVs to train the LSTM prediction models collaboratively without sharing raw user request data.
3. **Advanced Communication Technologies:** The integration of 6G and Terahertz (THz) communication models could be explored to further enhance data transmission rates and support ultra-reliable low-latency communications (URLLC) for safety-critical applications.
4. **Heterogeneous UAV Fleets:** Extending the framework to support a mix of different UAV types (e.g., rotary-wing for hovering, fixed-wing for coverage) with varying capabilities would better reflect real-world deployment scenarios.

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